

Solution Requirements

AI-ML & GEN-AI Track | Project Phase Documentation

Date	30 September 2026	Team ID	SWTID-2026-8540
Project Name	AI BlogNest API	Track	AI / ML / Generative AI
Team Size	5	Maximum Marks	4 Marks
Team Leader	Pravin kumar .P	Members	Abdul Rahaman .A; Dhamotharan .G; Maharaja .C; SUBHASH S

Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Secure registration/login and JWT-based stateless session handling.	High
2	Authorization	Role-based access for Admin, Editor, Author and Reader.	High
3	External Interfaces	Integrate Gemini AI through dedicated AI service endpoints.	High
4	Blog CRUD	Create, read, update and delete blog records with validation and status handling.	High
5	Comments	Create and moderate comments using protected routes.	Medium
6	Taxonomy & Search	Manage categories/tags and provide searchable content.	Medium
7	AI Assistance	Generate blog content and summarize long articles.	High
8	Media Metadata	Store media/path metadata associated with content.	Medium

Non-Functional Requirements (NFR)

S.No	Category	Target / Requirement
1	Security	Hash passwords with bcrypt.js; validate, sanitize and rate-limit requests.
2	Reliability	Centralized error handling and structured logging.
3	Maintainability	Layered modular architecture with separate models, controllers and routes.
4	Scalability	Support increased users/data through efficient database access and service separation.
5	Performance	Keep API responses predictable; avoid unnecessary database operations.